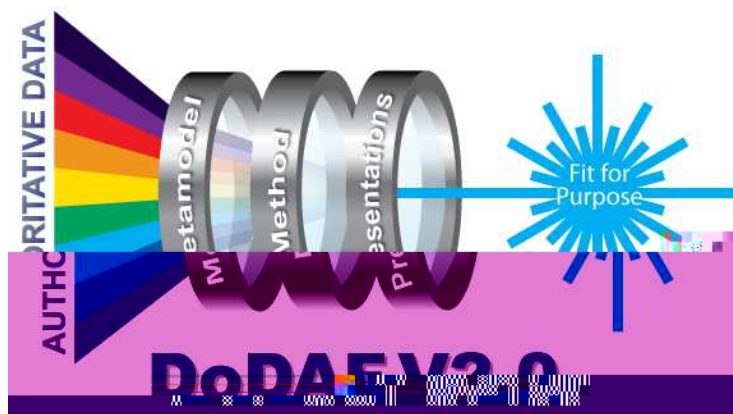


DRAFT



DoD Architecture Framework

Version 2.02, Change 1



Volume I: Overview and Concepts

Manager's Guide

31 January 2015

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Change 1 Description

DoDAF has had three incremental updates since the main release of DoDAF 2.0. Release 2.02, Chg 1 is primarily a result of the hard work and efforts of the Components in making refinements to existing content or from adding content provided by the architecture community on experimental techniques for organizing, sharing, and understanding the data in architectural descriptions. The United States Marine Corps (USMC) provided the predominant change through their revisions to the model descriptions in Volume II. Volume IV, from the Defense Chief Management Office (DCMO), provides new techniques and discussions for using DoDAF in conjunction with OWL and semantic web. Content location changes from Version 2.02 now keep information about specific topics in one place.

The DoDAF version 2.02, with its accompanying meta model, the DoDAF Meta Model (DM2), was baselined in October 2010. Improvements and corrections have been collected from the DoD EA community. These were logged and tracked by the DoDAF - DM2 Work Group (WG) secretariats following the processes and procedures documented in DoDAF - DM2 Configuration Management Plan. These were prioritized and adjudicated by the DoDAF - DM2 WG in a consensual manner through weekly DoDAF - DM2 WG meetings. WG actionees implemented the changes as per WG adjudication and reported back to the WG for WG review of the implementation.

There were 69 DoDAF-DM2 Action Items / Change Requests resolved by version 2.02, Chg 1.

- Information resource flow and associations were simplified into flatter type structure so it would be logically correct and consistent with other Resource Flows. (CR 642/652)
- Capability made a subtype of Property so that it is the set of Tasks performed under Conditions that meet certain performance standards (Measures). Also refined the Desired Effect of a Capability to be a Resource (state) that is desired by some PerformerCapableOfResponsibility. This makes Capability comparison and dependencies more direct as property intersections and Resource (state) overlaps. (CR 406/604, 453, 538, 598, 603, 605)
- Added SoA Joint Action concept and distinguished business services from enabling services (CR 597)
- Refined rules for superclass association usage (CR 503, 618)
- Continued work on refinement of meaning of Services. (CR 151)
- Relationship between Context and Condition clarified. (CR 91)
- Resources in LocationTypes and ResourceTypes in Locations clarified (CR 643)
- Several Data Dictionary and Alias corrections, e.g., Ways. (CR 414, 449/520, 549, 630)
- Several IDEAS Foundation corrections. (CR 295, 408, 439, 464, 484, 494, 497, 517, 541, 544, 548, 573, 595, 600, 606, 607, 609, 612, 619, 647)
- Several minor PES corrections (CR 405, 566a, 622, 641, 649)
- DoDAF website and FAQ improvements (CR 593, 402)

Executive Summary

The Department of Defense Architecture Framework (DoDAF) is the overarching, comprehensive framework and conceptual model for architectural descriptions developed within the DoD. This framework helps Department of Defense (DoD) managers at all levels make effective decisions by ensuring the sharing of consistent and common information across the Department, Joint Capability Areas (JCAs), missions, components, and programs. The DoDAF helps the DoD Chief Information Officer (CIO) develop and maintain architectures required by the Clinger-Cohen Act. It also fulfills guidance from the Office of Management and Budget (OMB) and other Departmental directives and instructions.

The DoDAF supports DoD's core decision-making processes, including the Joint Capabilities Integration and Development System (JCIDS), the Defense Acquisition System (DAS), Systems Engineering (SE), the Planning, Programming, Budgeting, and Execution (PPBE) Process, Capabilities Portfolio Management (CPM), and Operations (OPS).

DoD Components should conform to the DoDAF when they develop architectures within the Department.

The DoDAF allows architectural artifacts to be *fit-for-purpose*, that is, to be defined and described consistently with specific project or mission decision-making needs. Because architectural descriptions are employed at many levels, contexts, and purposes within the DoD, they vary in content, structure, and level of detail. Basing the architectural description development on well-articulated and understood purposes will ensure that the necessary data collection occurs at the appropriate level of detail to support specific decisions.

The DoDAF focuses on architectural data rather than architecture artifacts. It identifies, defines, and specifies the information needed to describe something in architectural terms within DoD. There is a wide range of architecture tools developed by commercial sources that can collect, organize, and store architecture data. The focus on data supports the production of fit-for-purpose models tailored for multiple uses. It also supports analysis and simulation of architectural description content produced across Components to support DoD's core decision making processes. Consequently, tools should use the DoDAF Meta Model (DM2) specifications to exchange architectural data.

Models visualize architecture data. A model, displayed as diagrams, narrative text, matrices, tables, dashboards, or other representations, serves as a template for organizing and displaying data in a format appropriate for a decision-maker. Viewpoints are thematic collections of models. A viewpoint focuses on data within the scope of some concern, such as capabilities, systems, or standards. A set of viewpoints, accompanied by useful definitions of the terms they use, is an architectural description.

The DoDAF specification comprises four volumes.

- Volume I, the manager's volume, provides general information and guidance for development, use, and management of DoD architectures. This volume explains the role of architecture within core DoD processes and key DoD architecture concepts are identified and defined.
- Volume II, the architect's volume: 1) defines architectural viewpoints and models, and 2) specifies the DM2 at a conceptual and logical level, through an elaboration of the key concepts. Models depict a subset of architectural data within a viewpoint. Once populated with data, models associated with the viewpoint can present these data. The DoDAF specifies over 50 standard models within eight viewpoints. The DM2 supports creating additional custom, fit for purpose, models to present architectural data within or across viewpoints for specific stakeholders and their specific needs.
- Volume III, the developer's volume, discusses the ontological foundation for DM2 and specifies the physical level format for the exchange of DoDAF-compliant architectural data. This volume is for developers of architectural description analytics, tools, databases, repositories, and simulations.
- Volume IV, the DoDAF Journal, publishes descriptions of other best practices, lessons learned, and reference documents that supplement the information contained in the three volumes of the DoDAF, including a discussion of the DoDAF OWL exchange specification. This volume provides information only and is not part of DoDAF conformance.

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1 INTRODUCTION

The Department of Defense Architecture Framework (DoDAF) is the overarching, comprehensive framework and conceptual model for architectural descriptions developed within the DoD. The DoDAF is the structure for organizing architecture concepts, principles, assumptions, and terminology about operations and solutions into meaningful and consistent patterns to satisfy specific DoD purposes. The DoDAF offers guidance, principles, and direction on communicating business and mission needs and capabilities to managers, architects, analysts, and developers who are responsible for developing and building the necessary systems, services, applications, and infrastructure to meet stakeholder needs and to manage their expectations.

This framework helps DoD managers at all levels make effective decisions by sharing information across the Department, Joint Capability Areas (JCAs), missions, components, and programs. The DoDAF focuses on the collection, presentation, and sharing of architectural data as information required by DoD decision makers, rather than on developing individual models. Architects may use the standard models described in this Volume I and specified in Volume II to obtain and visualize architecture data. However, the framework also allows architects to build other, fit-for-purpose (FFP) products for an architectural description.

1.1 Vision for the DoDAF

The vision for use of the DoDAF is to:

- Provide architecture concepts to guide development of architectures throughout the Department in support of decision processes for departmental programs, military components, and capability areas. This guidance is consistent with federal enterprise architecture guidance provided by OMB.
- Focus on architectural data as information required for making critical decisions and de-emphasize individual or independent architecture models. Allow architects to visualize architectural information using both standard models and fit-for-purpose models that are consistent with the culture and preferences of an organization while being consistent descriptions for consumption and use by the entire Department.

1.2 Purpose

The purposes of DoDAF are as follows.

- a. DoDAF supports the Department of Defense Chief Information Officer (DoD CIO) efforts to develop and maintain architectures as required by the Clinger-Cohen Act. From a compliance perspective, federal law and policy (i.e., Clinger-Cohen Act, OMB Circular A-130) require architectures to support investment decisions. The Office of Management and Budget (OMB) annually evaluates agency efforts to improve the quality and usefulness of information technology investments requested by agencies

through well-organized strategic decisions relating to investments and Portfolio Management. This process evaluates the use of enterprise architectures as the principal means of meeting mission requirements, while achieving savings and cost avoidance goals. Each agency is required to adopt an existing architecture framework or to create one for that purpose. The DoDAF is the designated architecture framework for DoD architecture development.

- b. DoDAF supports DoD's core decision-making processes, including the Joint Capabilities Integration and Development System (JCIDS), the Defense Acquisition System (DAS), Systems Engineering (SE), the Planning, Programming, Budgeting, and Execution (PPBE) Process, Capabilities Portfolio Management (CPM), and Operations (OPS). These key processes produce far-reaching change across all Military Departments, Agencies, the Joint Staff, and other Departmental functions.
- c. The framework is consistent with, and supports DoD policy directives that require programs and components to (a) ensure that their architectures meet stated objectives and departmental requirements, and, (b) provide the information necessary to support defined decisions at higher tiers. These policies also require consistency across horizontal architecture boundaries within a tier. The guidance and information contained in these volumes also ensures that, when followed, architecture development is consistent with OMB guidance on enterprise architecture.
- d. This version of the DoDAF supports the Departmental preference for federated architecture development in a tiered environment. To enable federation and support tiered responsibility and accountability, the framework provides data structures for comparing appropriate touch-points for consistency across architecture boundaries. Use of these data structures ensures that higher tiers have access to data from lower tiers in a form that supports their decision needs.
- e. Architecture frameworks support change in organizations through building and using architectures that:
 - Enhance decision making processes by leveraging knowledge in existing architectures and opportunities for reusing existing information assets.
 - Respond to stakeholder, customer, and client needs for effective and efficient processes, systems, services, and resource allocation.
 - Provide mechanisms to manage configuration of the current state of the enterprise and to maintain validity of the expected performance.
 - Analyze designs for future states of the enterprise.
 - Establish baseline architectures for solutions under development.

- f. From a practical perspective, an organization that pursues complex ends with sophisticated people, systems, services, and technologies needs comparably complex architectures to evaluate and compare investments. Such an organization also uses architectures to build new systems, deploy new technologies, offer new services, and guide change to the organization itself.
- g. The DoDAF also helps architects develop SOA-based a

2 DODAF VOLUME ORGANIZATION AND INTENDED AUDIENCE

The DoDAF has four volumes.

2.1 Volume I — Introduction, Overview, and Concepts

Primary audience: executives, project directors, and managers.

Volume I introduces DoD architecture concepts and provides general guidance for development, use, and management of DoD architectures. This volume explains the role of architecture within core DoD processes. Volume I identifies and defines key DoD architecture concepts.

Volume I contains the following resources:

- An overview and vision for DoDAF.
- An overview of the framework.
- Defining fit-for-purpose architectures.
- Introduction to the DoDAF Meta Model and identification and definition of key DoD architecture concepts.

2.2 Volume II — Architectural Data and Models

Primary audience: architects, program managers, portfolio managers, systems engineers, capability analysts and testers, and other technically oriented architecture users

Architects, modelers, and technical designers need to know what sorts of things can be modeled and the sorts of relationships among those things. Volume II describes the DoDAF meta-model, meta-model data groups, DoDAF viewpoints, and standard DoDAF models. The DoDAF meta-model specifies the sorts of things that can be modeled and the relationships among those things. Appendices to Volume II contain the DoDAF Glossary and references.

2.3 Volume III — DoDAF Meta Model Ontology Foundation and Physical Exchange Specification

Primary audience: developers of architectural description analytics, tools, databases, repositories, and simulations

Volume III discusses the ontological foundation for DM2 and specifies the physical level format for the exchange of DoDAF-compliant architectural data. These technical tools provide different ways to exchange architectural information among stakeholders.

2.4 Volume IV — DoDAF Journal

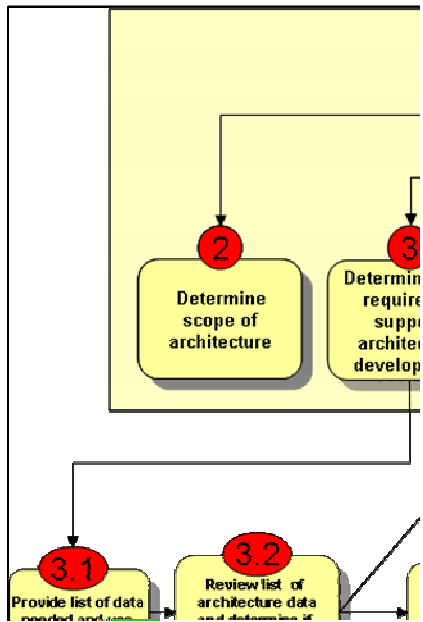
Volume IV is the informative volume of the DoDAF. Volume IV includes descriptions of best practices, lessons learned, reference documents, and other information that supplements the three normative volumes of the DoDAF.

3 WHAT DOD MANAGERS AND EXECUTIVES NEED TO KNOW ABOUT DODAF

Architecture development is a management tool that supports the decision-making process. A process owner, an executive responsible for a specific process or program, has the direct responsibility for ensuring that a particular process or program works efficiently, complies with legal and Departmental requirements, and serves the purpose for its creation. Legislation such as the Clinger-Cohen Act and implementing directives such as OMB Directive A-130 require periodic review and evaluation of the maturity and effectiveness of programs and processes. These requirements call for information architectures to support requests to fund those projects and processes.

A manager or executive may delegate the responsibility for creation of the architecture to a qualified architect working with an architecture development team. However, that delegation of authority does not alter the continuing responsibility of the executive or manager. As described throughout this volume, the decision-maker needs to be actively involved in the architecture development process and support architectural description development. They must also approve the architectural description for use and reference by the Department. Active involvement means that the decision-maker:

- Identifies the purpose and scope for the architecture.
- Transmits to the architect and development team the scope and purpose of the architecture effort, along with those goals and objectives that support the need.
- In conjunction with the architect, identifies the general data categories needed for architecture development, and assists in data collection and validation.
-



Figure

3.1 Developing Architectures

Careful scoping and organization of areas of change indicated by potential data-centric, rather than product-centric architectural models. Concordance when talking about the same expert architecture information across all models in an architectural design supports a wide variety of analysis. ‘property’ of architectures of all models in concordance across the architecture development actions.

The DoDAF describes three main architecture, Mission Area architecture, enterprise architectures. A fourth architecture but are not included in the DoDAF specific purpose:

- An enterprise architecture is a strategic information asset of an organization. This asset defines the mission of the organization, the behaviors and information necessary to perform the mission, the resources necessary to perform the mission, and the processes for transforming the organization and its resources to satisfy changing mission needs. An enterprise architecture includes a baseline architecture representing the current organization, a target architecture representing the future organization, and a plan for moving from the present into the future.
- Enterprise level reference architectures are an authoritative source of information about a specific subject area that guides and constrains the instantiations of multiple architectures and solutions. It has 5 elements:
 - Strategic Purpose – Identifies goals and objectives of the Reference Architecture and describes the specific purpose of and the problem(s) addressed by the Reference Architecture.
 - Principles – Sufficient high-level foundational statements of rules, culture, and values that drive technical positions and patterns.
 - Technical Positions– Technical guidance and standards, based on specified principles to follow and implement as part of the solution.
 - Patterns (Templates)– Generalized architecture representations (viewpoints, graphical/textual models, diagrams, etc.) that show relationships between elements and artifacts specified by the technical positions.
 - Vocabulary – Acronyms, terms, and definitions that are used in the Reference Architecture and relevant to architectures and solutions guided and constrained by the Reference Architecture.
- Component enterprise architectures are the description of mission-specific services and capabilities within the Component. It portrays relationships among all elements of a DoD Component.
- Solution architectures describe a system or other asset that an organization uses to carry out its mission. Although not part of the DoD enterprise architecture, solution managers use these architectures to create, update, revise, or remove resources that are called for by the organization's enterprise architecture. Solution architectures are the most common type of architecture developed in the Department.

3.2 Maintaining and Managing Architectures

Embedding architecture development process in routine planning and decision-making institutionalizes the practices of architecture and the maintenance of architectural data, models, and viewpoints. Tiered accountability provides the means to maintain and manage architectures

within the Department. Tiered accountability is the distribution of authority and responsibility for development, maintenance, configuration management, and reporting of architectures, architecture policy, tools, and related architecture artifacts to all three distinct tiers within the DoD. The DoDAF supports three tiers: Department, Mission Area (MA), and Component (i.e., enterprise and program or project-level solutions development). These tiers support the federated approach for architecture development and maintenance.

3.3 Using Architectures

Architecture supports major DoD decision-making processes, including JCIDS, DAS, PPBE, SE, and PfM processes. Architecture also supports business process reengineering, organizational development, research and development, operations support, and service-oriented solutions. Architectural data gives decision makers data they need to make informed decisions in those processes.

3.4 DoDAF Conformance

The Department of Defense expects DoD architectural descriptions to conform to the DoDAF to the maximum extent possible. Conformance ensures that reuse and sharing of information, architecture artifacts, models, and viewpoints is possible through a shared understanding of the underlying data. Both classified and unclassified architectural descriptions are to conform to the DoDAF.

There are four assessment level for DoDAF conformance. Higher levels of conformance build upon lower levels of conformance.

Level 1 — Conceptual conformance

- The architectural description uses normative DoDAF terms as defined in the DoDAF Glossary to identify concepts. The architectural description uses these normative DoDAF terms to describe the architecture. The AV-2 model, which is the glossary of the architectural description, appropriately defines additional terms used to describe the architecture. The AV-2 model complies with the DoDAF Glossary Style Manual guidance for writing definitions.
- DoDAF standard models within the architectural description satisfy the specifications given in Volume II.
- Stakeholders who use DoDAF fit-for-purpose models, validate them within the architectural description.

Level 2 — Logical conformance

- The architectural description demonstrates conceptual conformance.

- The AV-2 model within the architectural description complies with the DoDAF Glossary Style Manual guidance for constructing glossary entries and producing a glossary.
- The architectural description uses types, relationships, and properties defined by the DoDAF meta-model to describe the architecture. The architectural description correctly introduces and defines additional concepts, relationships, and properties used to describe the architecture as subtypes of DoDAF meta-model concepts, relationships, and properties.

Level 3 — Physical conformance

- The architectural description demonstrates logical conformance.
- The architectural data expressed by the architectural description is correctly produced and consumed using a specified format to exchange architectural data. A successful DM2 PES exchange satisfies this requirement; alternatively, architecture efforts within recognized Business Capability Lifecycle (BCL) programs may satisfy this criterion by successful DM2 OWL-DL exchanges.

Level 4 — Semantic conformance.

- The architectural description demonstrates physical conformance.
- The architectural description correctly uses and expresses the ontological semantics of the DoDAF meta-model.

4 DODAF STRUCTURE

The DoDAF's focus is on data, viewpoints, and models. This approach responds to departmental processes, such as business transformation, JCIDS, and other major functions with significant impact throughout the Department that have developed requirements for multiple, custom models. They use information based on authoritative data, beyond the operational, systems, and technical views of previous versions of DoDAF. The standard models are templates for identifying and collecting specific data within the data groups discussed in Volume II. Users define fit-for-purpose models to explain specific data to specific audiences.

4.1 Architectural Data

Architectural data provides efficient and flexible use and reuse of architectural descriptions for decision makers and process owners. The DoDAF metadata model (DM2) specifies a minimal set of architectural data required to support the core DoD decision-making processes. The DM2 has several levels, each of which is important to a particular viewer of departmental processes. The DoDAF conceptual data model (CDM) presents concepts shared by all DoDAF-compliant architectural descriptions. The CDM is defined in this Volume I, in this paragraph and in the Glossary in Appendix B.

The DoDAF logical data model (LDM) adds technical information and, when necessary, clarifies relationships into an unambiguous usage definition. Volume II discusses the LDM in detail.

DoDAF data exchange comes in two forms, the Physical Exchange Specification (PES) and the DM2-OWL specification. Volume III discusses the PES and Volume IV discusses DM2-OWL.

Data, organized as information, is the critical element of architecture development. The DoDAF provides the DM2 CDM, LDM, and the PES and OWL exchange specifications for data managers, tool vendors, and others to help:

- Establish areas of discourse and a shared vocabulary.
- Support data overlap analysis.
- Define and encourage the use of shared information.
- Provide a target for architectural data integration.

The DM2 defines architectural data elements and enables the integration and federation of architectural descriptions. It establishes a basis for semantic (i.e., understanding) consistency within and across architectural descriptions. In this manner, the DM2 supports the exchange and reuse of architectural information among MAs, Components, and federal and coalition partners; this helps the Department understand and build processes and systems that work well together, particularly in the sharing of information (interoperability).

4.1.1 The DoDAF Conceptual

The DoDAF conceptual data model is used to describe architectural descriptions. Key concepts may be read in a straightforward manner. The model is based on ovals and the predicate on the line



Figure 4-1. D

- An activity consumes and produces interesting resource. In general, an activity consumes resources. However, consumption is not always the case.
- An activity is performed by a performer.
- A performer is a sort of resource.

- An activity can produce a resource that performs another activity. Some activities, such as projects, are interesting just in that they produce performers that can realize capabilities.
- An activity is constrained by some guidance. Guidance forestalls random behavior. Proceeding by trial and error is not a best practice in anything we do.
- A rule is a sort of guidance.
- A standard is a sort of rule, and thus a standard is a sort of guidance.
- An agreement is a sort of rule, and thus an agreement is a sort of guidance.
- An activity is performed under some condition. Conditions affect the way a performer can carry out an activity, and conditions are seldom perfect in the real world.
- An activity is performed at some location. Locations are important for activities because they entail possible conditions.
- A resource exists at some location. Locations are important for resources because we cannot rely upon resources whose locations are unknown or unknowable.
- A geopolitical place is a sort of location.
- Materiel is a sort of resource. The DoDAF notion of materiel encompasses anything a performer uses to get a job done.
- A system is a sort of performer, and thus a system is a sort of resource.
- A service is a sort of performer, and thus a service is a sort of resource.
- An organization is a sort of performer, and thus an organization is a sort of resource.
- A person in a role is a sort of performer, and thus such a person is a resource.
- A performer can be a complex of systems, services, organizations, and persons in roles.
- A person in a role may be a part of a system.
- A person in a role may be a part of a service.
- A person in a role may be a part of an organization.
- Materiel may be a part of a performer.
- Information describes something. Specifically, information describes activities, guidance, conditions, resources, locations, and capabilities.
- Information is a sort of resource.

- Data is a sort of information, and thus data is a sort of resource. Data that is not used to describe activities, guidance, conditions, resources, locations, or capabilities is not architecturally interesting.

Further, the DoDAF conceptual data model inherits from the IDEAS ontology so that:

- Everything of architectural interest has four dimensions, that is, they exist in space and time. All the pieces and parts of a described architecture must be founded upon things that are real in the world.
- Everything of architectural interest has parts. In particular, everything has both temporal parts and spatial parts. This is the basis for asserting the identity of a whole as its parts change over time.
- Everything of architectural interest is a sort of something. Indeed, any given thing can be a sort of many different things at the same time and over time.
- Everything of architectural interest has measures. Something that exists in space and time can be observed. Anything that can be observed can be measured. At a minimum, we can measure the size and the position of any real thing of architectural interest.

Together, these concepts cover the notions needed to discuss all aspects of architectural description in DoD. For example, *capabilities* as defined by Joint doctrine:

A capability is the ability to achieve a desired effect under specified standards and conditions through combinations of means and ways to perform a set of tasks.

A desired effect is a measurable change in the state of resources we see in someplace in the world. Activities consume resources in one state and produce resources in another state. Performers perform activities that change the state of resources. Performers do this under conditions that affect their performance. Performers do this following guidance to perform tasks appropriately under those conditions. All this can be measured, and the performance of an activity can be assessed against standards of performance. In architectural terms: *tasks* are activities, *ways* are guidance, *means* are performers, *conditions* are conditions, *standards* are a particular sort of guidance, and *desired effects* are changes in the states of resources.

In the DM2 these straightforward concepts are founded on a formal ontology that enables architectural descriptions as complex and detailed as required. Figure 4-2 illustrates key concepts of the DM2's foundation.

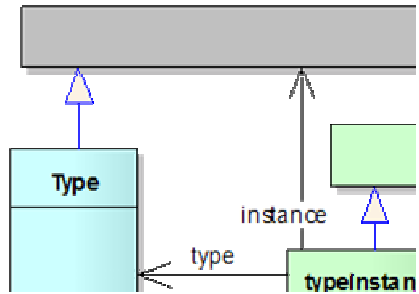


Figure 4-2.

The top-level foundation elements are:

- thing — anything that is represented in the model.
- individual — a thing that is an instance of a type.
- type — a grouping of things that share common characteristics.
- tuple — an ordered pair of things.

The foundation tuples (relationships) are:

- whole & part — composition relationships.
- supertype & subtype — generalization relationships.
- before & after — temporal relationships.
- overlap — four-dimensional relationships.

Composition and specialization are used to arrange things through time. (These are not necessarily contained within each other.)

4.2 Architecture Viewpoints and DoDAF-described Models

An architecture viewpoint is a selected set of architectural data organized around some central concept. There are many ways to present an architectural description. A model, regardless of its form, is a representation of some portion of the architectural data, in the sense that a still photograph shows only one view of a subject within a picture. Figure 4-3 provides a graphical representation of the architecture viewpoints in the DoDAF.

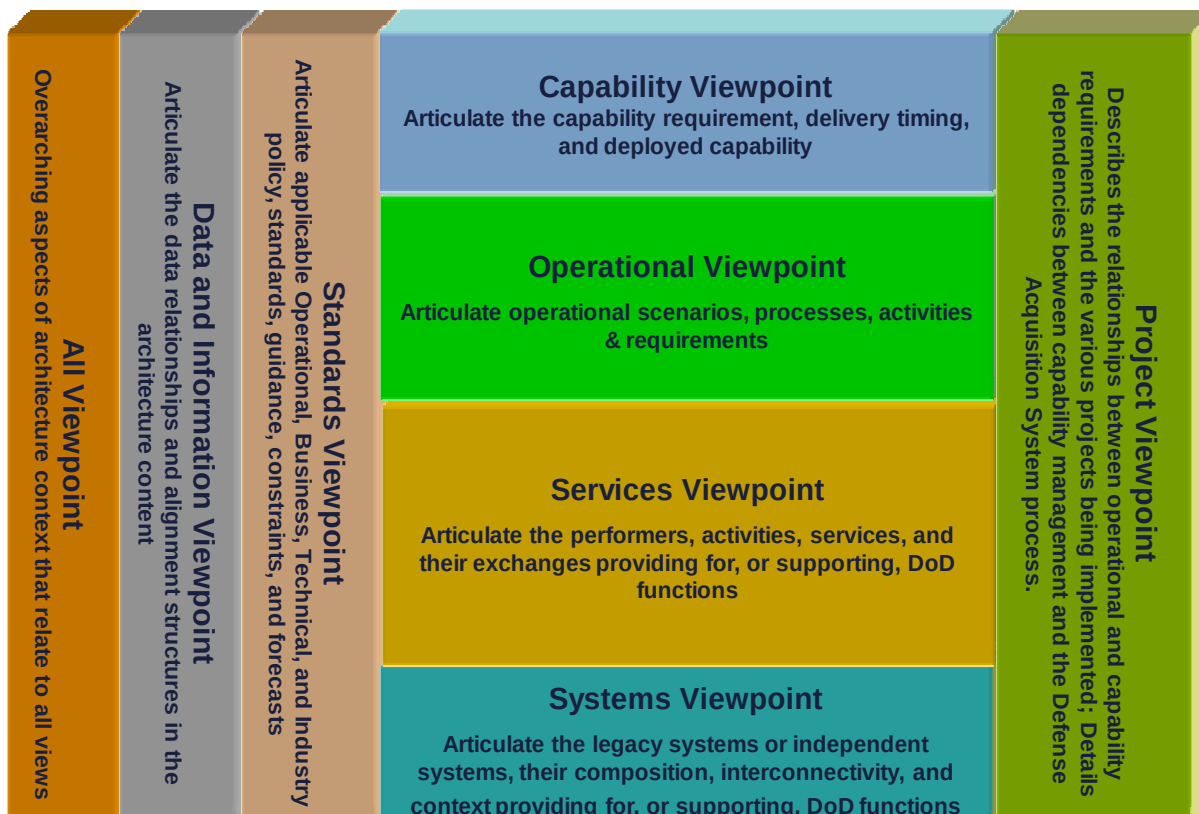


Figure 4-3. Architecture Viewpoints in the DoDAF

4.2.1 All Viewpoint

Some overarching aspects of an architectural description relate to all models. The All Viewpoint (AV) models provide information about the entire architectural description, such as its scope and context. The scope includes the subject area and timeframe of the architectural description. The setting in which the architectural description exists comprises the interrelated conditions that compose the context for the architectural description. These conditions include doctrine; tactics, techniques, and procedures; relevant goals and vision statements; concepts of operations (CONOPS); scenarios; and environmental conditions.

4.2.2 Capability Viewpoint

The Capability Viewpoint (CV) describes a vision for performing specified activities to achieve desired resource states under specified standards and conditions. It applies specified guidance and specified performers to those tasks. This viewpoint provides a strategic rationale for the described architecture.

4.2.3 Data and Information Viewpoint

The Data and Information Viewpoint (DIV) describes information needs, data requirements, and the implementation of data elements within an architectural description. This viewpoint includes information associated with information exchanges in the architectural description, such as the attributes, characteristics, and inter-relationships of exchanged data.

4.2.4 Operational Viewpoint

The Operational Viewpoint (OV) describes organizations, activities they perform, and resources they exchange to fulfill DoD missions. This viewpoint includes the types of information exchanged, the frequency of such exchanges, the activities supported by information exchanges, and the nature of information exchanges.

4.2.5 Project Viewpoint

The Project Viewpoint (PV) describes how programs are grouped in organizational terms as a coherent portfolio of acquisition programs. This viewpoint provides a way of describing the organizational relationships between multiple acquisition programs, each of which is responsible for delivering systems or capabilities.

4.2.6 Services Viewpoint

The Services Viewpoint (SvcV) describes services that provide or support operational activities. This viewpoint traces service activities and resources to the requirements established by the Operational Viewpoint.

4.2.7 Standards Viewpoint

The Standards Viewpoint (StdV) describes the minimal set of rules governing the arrangement, interaction, and interdependence of systems and system parts. The purpose of this viewpoint is to ensure that a system satisfies a specified set of operational requirements. The Standards Viewpoint identifies the technical systems implementation guidelines upon which engineering specifications are based, common building blocks established, and product lines developed. This viewpoint includes a collection of the technical standards, implementation conventions,

standards options, rules, and criteria for organizing them into profiles that govern systems and system or service elements in a given architectural description.

4.2.8 Systems Viewpoint

Systems Viewpoint (SV) describes system activities and resources that support operational activities. This viewpoint traces system activities and resources to the requirements established by the Operational Viewpoint.

4.2.9 Standard Models

The table, DoDAF Standard Models, list the standard models provided by the DoDAF for the eight DoDAF viewpoints.

Table 4-1. DoDAF Standard Models

Model	Describes...
AV-1: Executive Summary	Project visions, goals, objectives, plans, activities, events, conditions, measures, effects (outcomes), and produced objects.
AV-2: Glossary	Definitions of ontic terms used in an architectural description.
CV-1: Capability Effects	The overall vision for transformational endeavors, which provides a strategic context for the capabilities described and a high-level scope.
CV-2: Capability Hierarchies	hierarchy of capabilities which specifies all the capabilities that are referenced throughout one or more architectural descriptions.
CV-3: Capability Schedules	The planned achievement of capability at different points in time or during specific periods of time. The "#-\$ shows the capability phasing in terms of the activities, conditions, desired effects, rules complied with, resource consumption and production, and measures, without regard to the performer and location solutions.
CV-4: Capability Dependencies	The dependencies between planned capabilities and the definition of logical groupings of capabilities.
CV-5: Capability Deployments	The fulfillment of capability requirements shows the planned capability deployment and interconnection for a particular capability phase. The "#-& shows the planned solution for the phase in terms of performers and locations and their associated concepts.
CV-6: Capability Activities	mapping between the capabilities required and the operational activities that those capabilities support.
CV-7: Capability & Services	mapping between the capabilities and the services that these capabilities enable.
DIV-1: Conceptual Information	Information needs.
DIV-2: Data Requirements Model	Data requirements.
DIV-3: Data Implementation	The physical implementation of data elements.
OV-1: Operational Concept	The operational concept.
OV-2: Organizations & Resources	(resource flows exchanged between operational activities.
OV-3: Organizations, Activities, & Resources	(resources exchanged and the relevant attributes of the exchanges.
OV-4: Organizational Relationships	Organizational context, roles, and other relationships among organizations.
OV-5a: Operational Activity Hierarchy	Capabilities and operational activities organized in a hierarchical structure.
OV-5b: Operational Activities	The context of capabilities and operational activities and the relationships among activities, inputs, and outputs.
OV-6a: Operational Rules	Rules that constrain operational activities.
OV-6b: Operational State Transitions	Activity responses to other activities.
OV-6c: Operational Activity Sequences	Activities in a scenario, a specified sequence of activities.
PV-1: Projects & Organizations	The dependency relationships between the organizations and projects and the organizational structures needed to manage a portfolio of projects.
PV-2: Project Schedules	schedule of activities and their resources with the key milestones and dependencies.
PV-3: Projects & Capabilities	mapping of programs and projects to capabilities to show how the specific projects and program elements help to achieve a

Model	Describes...
	capabilities.
SvcV-1 Services	services, service items, and their interconnections.
SvcV-2 Services Interfaces	resource flows among services.
SvcV-3a Services & Systems	relationships among or between systems and services in a given architectural description.
SvcV-3b Service Relationships	relationships among services in a given architectural description.
SvcV-4 Services Functions	activities performed by services and the service resource flows among service activities.
SvcV-5 Services & Operational Activities	mapping of service activities to operational activities.
SvcV-6 Services, Activities, & Resources	service resource flow among between services and the attributes of those resources.
SvcV-7 Service Measures	measures of services for interesting periods of activity.
SvcV-8 Services Evolution	Planned incremental steps to migrate from current services to future services.
SvcV-9 Service Technologies & Skills	merging resources, standards, and skills that planners expect to be available for future service development.
SvcV-10a Services Rules	rules that constrain service activities.
SvcV-10b Services State Transitions	service activity responses to other activities.
SvcV-10c Services Activity Sequences	activities in a scenario, a specified sequence of service activities.
StdV-1 Standards Profile	current standards constraining activities that produce solution resources.
StdV-2 Standards Forecast	future standards that will constrain activities that produce solution resources.
SV-1 Systems Composition and Interface Identification	systems, system parts, and their relationships.
SV-2 System Interface Means	resource flows among systems.
SV-3 System Relationships	relationships among systems in an architectural description.
SV-4 Systems Functions	The functions (activities) performed by systems and the system data flows among system functions (activities).
SV-5a Systems & Operational Activities	The relationships of system activities to operational activities.
SV-5b Systems & Capabilities	mapping of systems back to capabilities or operational activities (activities).
SV-6 Systems, Activities, & Resources	Provides details of system resource flow elements being exchanged between systems and the attributes of that exchange.
SV-7 System Measures	measures of a system.
SV-8 System Evolution	The plan to upgrade a suite of systems to a more efficient suite or to evolve a current system to a future implementation.
SV-9 System Technologies & Skills	The emerging technologies, software/hardware products, and skills that are expected to be available in a given set of time frames and that will affect future system development.
SV-10a Systems Rules	constraints on system activities.
SV-10b System State Transitions	how a system responds to events.
SV-10c System Activity Sequences	system-specific refinements of critical sequences of activities described in the operational viewpoint.

APPENDIX A ACRONYMS & ABBREVIATIONS

Acronym	Definition
AV	All Viewpoint
BEA	Business Enterprise Architecture
BMA	Business Mission Area
BPMN	Business Process Modeling Notation
C2	Command and Control
CA	Common Approach
CDM	Conceptual Data Model
CI	Configuration Item
CM	Configuration Management
COI	Community Of Interest
COMMPLAN	Communications Plan
CDD	Capability Development Document
CPD	Capability Production Document
CPM	Capability Portfolio Management
CV	Capability Viewpoint
DAS	Defense Acquisition System
DDMS	Department of Defense Discovery Metadata Specification
DISR	DoD Information Standards Registry
DIV	Data and Information Viewpoint
DM2	DoDAF meta-model
DNDAF	Department of National Defense Architecture Framework
DOTMLPF	Doctrine, Organization, Training, Material, Leadership and education, Personnel, and Facilities
E-R	Entity-Relationship
EA	Enterprise Architecture
EEI	Essential Element of Information

Acronym	Definition
FEA	Federal Enterprise Architecture
FFP	Fit For Purpose
FOC	Full Operational Capability
IC	Intelligence Community
IC-ISM	Intelligence Community – Intelligence Standard Markings
ICD	Initial Capabilities Document
IDEAS	International Defence Enterprise Architecture Specification
IEA	Information Environment Architecture
IER	Information Exchange Requirement
IMA	Information Mission Area
IPB	Intelligence Preparation of the Battlefield
IPOE	Intelligence Preparation of the Operational Environment
IOC	Initial Operational Capability
ISO	International Standards Organization
ISP	Interoperability Support Plan
ISR	Intelligence, Surveillance and Reconnaissance
IT	Information Technology
JCA	Joint Capability Areas
JCIDS	Joint Capabilities Integration and Development System
LDM	Logical Data Model
OMB	Office of Management and Budget
OPLAN	Operation Plan
OV	Operational Viewpoint
MODAF	Ministry of Defence Architecture Framework
MOE	Measure Of Effectiveness
MOP	Measure of Performance
NIEM	National Information Exchange Model

Acronym	Definition
NSS	National Security System
PE	Program Element
PES	Physical Exchange Specification
PIR	Priority Intelligence Requirement
POM	Program Objective Memorandum
PPBE	Planning, Programming, Budgeting, and Execution
PV	Project Viewpoint
QoS	Quality of Service
RA	Reference Architecture
RDBMS	Relational Database Management System
SA	Solution Architecture
SCI	Software Configuration Item
SE	Systems Engineering
SETR	System Engineering Technical Review
SOA	Service Oriented Architecture
SoS	System of Systems
SoSE	System of Systems Engineering
SV	Systems Viewpoint
SvcV	Services Viewpoint
StdV	Standards Viewpoint
TADIL	TActial Data and Information Link
TEMP	Test and Evaluation Master Plan
TOGAF	The Open Group Architecture Framework
TTP	Tactics, Techniques, and Procedures
UJTL	Universal Joint Task List
UML	Unified Modeling Language
URL	Universal Resource Locator

Acronym	Definition
WBS	Work Breakdown Structure
WMA	Warfighting Mission Area
XML	Extensible Markup Language
XSD	XML Schema Definition

APPENDIX B GLOSSARY OF CONCEPTUAL LEVEL TERMS

This appendix lists conceptual level terms and their definitions from the DoDAF Glossary¹. For more information about these terms, including their technical DM2 definitions, consult the complete Glossary. These definitions are generally stated in the singular; however, this grammar assumes that whatever applies to one also applies to many. Consult WordNet for the meaning of terms not defined here. The appropriate senses among those given by WordNet are noted by an index number in entries that specify a specific sense of term for DoDAF use.

activity — a transformation of some resource into another resource.

agreement — a guidance statement that records consent among performers to guidance and conditions for performing an activity.

capability — an ability to achieve a desired resource state under a specified performance standard and a specified condition through some combination of guidance and resources to perform a set of activities. ♦ *translated from*: Joint Publication X.

condition — a state of resources that affects the performance of an activity.

data — an information resource that represents states in a standard way suitable for consumption and production by activities. • *see*: information.

desired resource state — a state of resources that is envisioned by a performer capable of responsibility. • *see*: vision, capability, resource. • *note*: A desired resource state is the DoDAF expression of the desired effect of a capability. In the Joint view of capability, a performer capable of responsibility is exemplified by a combatant commander.

geopolitical extent — a region of the world whose boundaries are asserted by a nation state.

guidance — an information resource that is an authoritative statement that constrains the performance of an activity.

information — a resource that is a representation of the state of rules, conditions, activities, performers, and other resources. • *note*: Information is often produced by one performer to be consumed by another, decision-making performer. • *example*: *Information is a difference that makes a difference.* • Gregory Bateson.

location — a point or extent in space that may be referred to by coordinates or by name. • *note*: A location is said to be a *geospatial extent*.

¹ The DoDAF 2.02, Chg 1 Glossary is also known as the DoDAF 2.02, Chg 1 Data Dictionary.

materiel — a resource that is some assemblage of equipment, apparatus, and supplies used by a performer to perform an activity.

measure — a quantification of the magnitude of some property of a thing.

organization — a performer that is an assemblage of persons in roles and resources that support those roles.

performer — a resource that performs an activity.

performer capable of responsibility — a person in a role that is accountable for the performance of an activity. • *see*: person role.

person role — a performer that is a person defined by a role with respect to an activity. • *note*: In day-to-day language, we speak of a *person in a role*.

resource — any thing that is produced or consumed by an activity. • *note*: Performers and guidance associated with an activity are themselves products of other activities.

rule — a guidance statement that prescribes the performance of an activity.

service — a performer that enables access to the performance of a set of activities.

standard — a guidance statement that specifies criteria for the performance of an activity.

system — a performer that is an assemblage of resources.

vision — an information resource that describes a future state of resources that is to be achieved.

APPENDIX C RECORD OF CHANGES FOR CHANGE 1

#	Title	Description	Source Org	Action

#	Title	Description
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#	Title	Description	Source Org	Action

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